

CALCULATED FIELDS WITH DAX

BASIC MATH & STATS FUNCTIONS

SUM	Evaluates the sum of a column	= SUM (ColumnName)
AVERAGE	Returns the average (arithmetic mean) of all the numbers in a column	= AVERAGE (ColumnName)
MAX	Returns the largest value in a column or between two scalar expressions	= MAX (ColumnNameOrScalar1, [Scalar2])
MIN	Returns the smallest value in a column or between two scalar expressions	= MIN (ColumnNameOrScalar1, [Scalar2])
DIVIDE	Performs division and returns the alternate result (or blank) if DIV/0	= DIVIDE (Numerator, Denominator, [AlternateResult])

COUNTING FUNCTIONS

COUNT	Counts the number of non-empty cells in a column (excluding Boolean values)	= COUNT (ColumnName)
COUNTA	Counts the number of non-empty cells in a column (including Boolean values)	= COUNTA (ColumnName)
DISTINCTCOUNT	Counts the number of distinct values in a column	= DISTINCTCOUNT (ColumnName)
COUNTROWS	Counts the number of rows in the specified table, or a table defined by an expression	= COUNTROWS ([Table])

Example:

```
1 Total Customers =  
2 DISTINCTCOUNT(  
3 | 'Sales Data'[Customer Key]  
4 )
```

```
1 Return Rate =  
2 DIVIDE(  
3 | [Quantity Returned],  
4 | [Quantity Sold],  
5 | "No Sales"  
6 )
```

```
1 Total Returns =  
2 COUNTROWS(  
3 | "Returns Data"  
4 )
```

BASIC LOGICAL FUNCTIONS

IF	Checks if a given condition is met and returns one value if the condition is TRUE, and another if the condition is FALSE	= IF (LogicalTest, ResultIfTrue, [ResultIfFalse])	
IFERROR	Evaluates an expression and returns a specified value if it returns an error, otherwise returns the expression itself	= IFERROR (Value, ValueIfError)	
SWITCH	Evaluates an expression against a list of values and returns one of multiple possible expressions	= SWITCH (Expression, Value1, Result1, ..., [Else])	
AND	Checks whether both arguments are TRUE to return TRUE, otherwise returns FALSE	= AND (Logical1, Logical2)	} Note: Use the && and operators to include more than two conditions
OR	Checks whether any argument is TRUE to return TRUE, otherwise returns FALSE	= OR (Logical1, Logical2)	

SWITCH

SWITCH

Evaluates an expression against a list of values and returns one of multiple possible expressions

=**SWITCH**(Expression, Value1, Result1, ..., [Else])

Any **DAX expression** that returns a single scalar value, evaluated multiples times

Examples:

- Calendar[Month ID]
- 'Product Lookup'[category]

List of **values** produced by the expression, each paired with a result to return for rows/cases that match

Examples:

=**SWITCH**(Calendar[Month ID],
1, "January",
2, "February"

Value returned if the expression doesn't match any value argument



PRO TIP

SWITCH(TRUE) is a common DAX pattern to replace multiple nested IF statements

Example:

```
1 Customer Priority =  
2 IF(  
3     'Customer Lookup'[AnnualIncome] > 100000 &&  
4     'Customer Lookup'[Is Parent?] = "Yes",  
5     "Priority",  
6     "Standard"  
7 )
```

```
1 Income Level =  
2 IF('Customer Lookup'[AnnualIncome] >= 150000, "Very High",  
3 IF('Customer Lookup'[AnnualIncome] >= 100000, "High",  
4 IF('Customer Lookup'[AnnualIncome] >= 50000, "Average",  
5 "Low"))))
```

```
1 Education Category =  
2 SWITCH('Customer Lookup'[EducationLevel],  
3     "High School", "High School",  
4     "Partial High School", "High School",  
5     "Bachelors", "Undergrad",  
6     "Partial College", "Undergrad",  
7     "Graduate Degree", "Graduate")
```

TEXT FUNCTIONS

LEN	Returns the number of characters in a string	=LEN(Text)	Note: Use the & operator as a shortcut, or to combine more than two strings
CONCATENATE	Joins two text strings into one	=CONCATENATE(Text1, Text2)	
UPPER/LOWER	Converts a string to upper or lower case	=UPPER/LOWER (Text)	
LEFT/RIGHT/MID	Returns a number of characters from the start/middle/end of a text string	=LEFT/RIGHT(Text, [NumChars]) =MID(Text, StartPosition, NumChars)	
SUBSTITUTE	Replaces an instance of existing text with new text in a string	=SUBSTITUTE(Text, OldText, NewText, [InstanceNumber])	
SEARCH	Returns the position where a specified string or character is found, reading left to right	=SEARCH(FindText, WithinText, [StartPosition], [NotFoundValue])	

Example:

```
1 Month Short =  
2 UPPER(  
3     LEFT(  
4         'Calendar Lookup'[Month Name],  
5         3  
6     )  
7 )
```

```

1 SKU Category =
2 LEFT(
3     'Product Lookup'[Product SKU],
4     SEARCH(
5         "-",
6         'Product Lookup'[Product SKU]
7     )
8     -1
9 )

```

BASIC DATE & TIME FUNCTIONS

TODAY/NOW	Returns the current date or exact time	= TODAY/NOW ()
DAY/MONTH/YEAR	Returns the day of the month (1-31), month of the year (1-12), or year of a given date	= DAY/MONTH/YEAR (Date)
HOUR/MINUTE/SECOND	Returns the hour (0-23), minute (0-59), or second (0-59) of a given datetime value	= HOUR/MINUTE/SECOND (Datetime)
WEEKDAY/WEEKNUM	Returns a weekday number from 1 (Sunday) to 7 (Saturday), or the week # of the year	= WEEKDAY/WEEKNUM (Date, [ReturnType])
EOMONTH	Returns the date of the last day of the month, +/- a specified number of months	= EOMONTH (StartDate, Months)
DATEDIFF	Returns the difference between two dates, based on a given interval (day, hour, year, etc.)	= DATEDIFF (Date1, Date2, Interval)

Example:

```
1 Birth Year =  
2 YEAR(  
3 | 'Customer Lookup'[BirthDate]  
4 )
```

RELATED

RELATED()

Returns related values in each row of a table based on relationships with other tables

=**RELATED**(ColumnName)

The **column** from a related table containing the values you want to retrieve

Examples:

- 'Product Lookup'[Product Name]
- 'Territory Lookup'[Country]

HEY THIS IS IMPORTANT!

RELATED works like a **VLOOKUP** function in Excel – it uses the relationship between tables (*defined by primary and foreign keys*) to pull values from one table into a new column of another.

Since this function requires row context, it can only be used as a **calculated column** or as part of an **iterator function** that cycles through all rows in a table (*FILTER, SUMX, MAXX, etc.*)



PRO TIP:

Instead of using RELATED to create extra columns (which increases file size), **nest it within measures like FILTER or SUMX**